Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [12CE8403]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

GREEN TECHNOLOGY AND SUSTAINABILITY

Time: 3 Hours Maximum Marks: 70

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	What are the different Renewable and Non-Renewable energy sources in India? Explain any two of them.	08
	b.	In how many ways our energy system can be made more earth friendly and sustainable?	06
2.	a.	Explain the principle of conversion of solar energy into heat.	06
	b.	Write a short note on multi-tracking facility for a solar power plant.	05
	c.	Name the locations where solar and wind power development in Karnataka and Gujarat state of India are located.	03
3.	a.	What do you understand by the term "Carbon footprint"? Explain with an example.	04
	b.	Comment on the following statement "Will voluntary carbon offsets alone solve climate change problem?".	04
	c.	How is environment affected by growing population? Explain with examples.	06
4.	a.	What is green cement and why it is gaining popularity in construction industry?	04
	b.	List various types of smart building materials and mention their advantages.	05
	c.	Explain the functioning of polyurethane water proofing system with a neat sketch.	05
5.	a.	Explain the heat transfer mechanism in building components.	07
	b.	How "Green technology Initiatives" can be planned to minimize the pollution from thermal power plants in India?	07
6.	a.	Discuss energy efficient high performance window construction practices in a green building.	06
	b.	Write a short note on green building rating system.	04
	c.	Define energy efficient landscaping for a building.	04
7.	a.	Would sensor technology help to reduce the risk of theft and fire in a modern residential building? Justify your statement.	07
	b.	Briefly explain the basic principle and application of fiber optics technology.	07
8.	a.	What are the basic components of a rainwater harvesting structure? Explain with a neat sketch.	08
	b.	In how many ways you can reuse the harvested rain water of a building?	03
	c.	How is the pipe diameter for tapping the water from roof top of a building arrived at?	03